

Interpretation & Explanation

Descriptive Statistics

The mean sales value represents the average sales generated across all orders. The median sales value indicates the middle sales amount when all sales are arranged in ascending order. The mode represents the most frequently occurring sales value. Variance and standard deviation measure the spread of sales values around the average. A higher standard deviation indicates greater variability in sales performance.

Probability Analysis

The probability calculation shows the likelihood of an order having total sales greater than ₹5000. For example, if the probability is 0.40, it means that 40% of all orders generated sales above ₹5000. This helps identify how frequently high-value sales occur.

Confidence Interval

The 95% confidence interval provides a range within which the true average sales of the population is expected to lie. For example, if the interval is ₹5800 to ₹6700, we can be 95% confident that the actual average sales falls within this range.

One-Sample t-Test

The one-sample t-test was used to determine whether the average sales significantly differ from the benchmark value of ₹5000. If the p-value is less than 0.05, the null hypothesis is rejected, indicating that the average sales are significantly different from ₹5000. If the p-value is greater than 0.05, there is insufficient evidence to conclude that the average sales differ from ₹5000.

Chi-Square Test

The Chi-Square test was conducted to examine whether there is a relationship between sales region and product category. A p-value less than 0.05 indicates a significant association between the two variables, while a p-value greater than 0.05 suggests that they are independent.

Histogram Interpretation

The histogram displays the distribution of total sales values. It helps identify whether sales are concentrated around a particular range and whether the distribution is symmetric or skewed.

Boxplot Interpretation

The boxplot summarizes the spread of sales values and highlights any outliers. Outliers represent unusually high or low sales transactions that may require further investigation.

Region-wise Sales Analysis

The bar chart compares total sales across different regions. Regions with higher sales indicate stronger market performance and may deserve additional business focus.

Category-wise Sales Analysis

The pie chart shows the contribution of each product category to total sales. Categories with larger percentages contribute more to overall revenue and represent important business segments.

Overall Conclusion

The statistical analysis provides valuable insights into sales performance through descriptive and inferential techniques. The findings help identify sales trends, variability, regional performance, and category contributions, enabling better business decisions and future planning.